PROBLEM 43

Classic Central Tether Problem

Two Connected Particles, Plane with Hole

Original statement · Problems.pdf, p. 30

We consider a smooth horizontal plane with a small hole in it. One of the ends of a massless cord of length $2a$ is passed through the hole and then, two point particles, P_1 and P_2, each with the same mass m, are attached to the ends of the cord. Therefore, P_1 is constrained to move on the plane and P_2 moves along a vertical line below the hole. Initially, half of the cord lies under the plane and half of the cord is on the plane. At that instant, P_1 is pushed with a speed v_0 perpendicular to the cord, counterclockwise as seen from above. Assume that the cord is always in tension.

(a) What is the value of v_0 if P_1 describes a circumference?

(b) In that case, what is the value of the tension in the cord?

(c) If $v_0 = \sqrt{8ag/3}$, reduce the problem to integrals (i.e., quadratures).

(d) Show that for this value of v_0, P_1 oscillates between $r = a$ and $r = 2a$.

(e) Apply Newton's law to P_2 and show that the tension of the cord is $T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$.

(f) Write down the new equations of motion if the plane on which P_1 moves is rough, with a friction coefficient μ.

(Hint: the roots of the polynomial $-3x^3 + 7x^2 - 4$ are $x_1 = 1, x_2 = 2$ and $x_3 = -2/3$).

Full Analytical Solution (4-Phase Methodology)

PHASE 1

Physical Statement, Hypotheses & Parameters

- **Physical System:** Two particles P_1 and P_2, each of mass m, connected by a massless cord of length $2a$.
- Cord passes through a smooth hole at origin O.
- P_1 moves on a smooth horizontal plane Oxy at distance $r(t)$.
- P_2 moves vertically below the hole at depth $z_2 = -(2a - r)$.
- **Initial State ($t = 0$):** $r(0) = a$, $z_2(0) = -a$. Pushed perpendicularly with v_0.
- **CDOFs:** 2 degrees of freedom: (r, θ).
- **Parameters:** $m, a, g, v_0 > 0$.

PHASE 2

Reference Frames, Vector Bases & Coordinate Geometry

- Polar coordinates for P_1: $\mathbf{r}_1 = r\mathbf{e}_r$, $\mathbf{v}_1 = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta$.
- Vertical position of P_2: $\mathbf{r}_2 = -(2a - r)\mathbf{k}$, $\mathbf{v}_2 = \dot{r}\mathbf{k}$.
- Total kinetic energy:

$$T = \frac{1}{2}mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + \frac{1}{2}m\dot{r}^2 = m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2$$

- **Change of basis (polar basis of P_1).** $\mathcal{B}_1 = \{\mathbf{e}_r, \mathbf{e}_\theta, \mathbf{k}\}$ is the inertial basis $\{\mathbf{i}, \mathbf{j}, \mathbf{k}\}$ rotated by $\theta(t)$ about $\mathbf{k}$:

$$[{}_0R_1] = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \qquad \det[{}_0R_1] = \cos^2\theta + \sin^2\theta = 1, \qquad [{}_0R_1][{}_0R_1]^T = I$$

- With $\boldsymbol{\omega}_{10} = \dot{\theta}\,\mathbf{k}$, Poisson's formula gives $\dot{\mathbf{e}}_r = \dot{\theta}\,\mathbf{e}_\theta$ and $\dot{\mathbf{e}}_\theta = -\dot{\theta}\,\mathbf{e}_r$, which produce the velocity $\mathbf{v}_1$ above. The matrix was checked symbolically with sympy.

PHASE 3

Step-by-Step Mathematical Deduction

Part (a)-(b): Circular Orbit Condition & String Tension

For P_1 to describe a circle of radius $r = a$: $\dot{r} = 0, \ddot{r} = 0$.

- For P_2 to remain suspended in equilibrium:

$$\sum F_z = T - mg = 0 \implies T = mg$$

- For P_1 in circular orbit: the tension provides the centripetal force:

$$T = m\frac{v_0^2}{a} \implies mg = m\frac{v_0^2}{a} \implies v_0 = \sqrt{ga}$$

String tension: $T = mg$.

Part (c): Reduction to Quadratures for $v_0 = \sqrt{\frac{8ga}{3}}$

1. **Angular Momentum Conservation for P_1:**

Tension on P_1 is purely radial ($\mathbf{T}_1 = -T\mathbf{e}_r$). No torque about O:

$$L_z = mr^2\dot{\theta} = mav_0 \implies \dot{\theta} = \frac{av_0}{r^2}$$

2. **Mechanical Energy Conservation:**

$$E = m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + mgz_2 = m\dot{r}^2 + \frac{ma^2v_0^2}{2r^2} - mg(2a - r)$$

Evaluate E at $t = 0$ with $r(0) = a, \dot{r}(0) = 0, v_0^2 = \frac{8ga}{3}$:

$$E = 0 + \frac{1}{2}mv_0^2 - mga = \frac{1}{2}m\left(\frac{8ga}{3}\right) - mga = \frac{4}{3}mga - mga = \frac{1}{3}mga$$

Energy equation:

$$m\dot{r}^2 + \frac{ma^2(8ga/3)}{2r^2} + mgr - 2mga = \frac{1}{3}mga$$

$$m\dot{r}^2 + \frac{4mga^3}{3r^2} + mgr - \frac{7}{3}mga = 0$$

Divide by m and isolate $\dot{r}^2$:

$$\dot{r}^2 = \frac{7}{3}ga - gr - \frac{4ga^3}{3r^2} = \frac{g}{3r^2}\left(-3r^3 + 7ar^2 - 4a^3\right)$$

Separating variables:

$$\int_a^{r(t)} \frac{r'\,dr'}{\sqrt{-3r'^3 + 7ar'^2 - 4a^3}} = \pm\sqrt{\frac{g}{3}}\,t$$

Part (d): Proof that P_1 Oscillates between $r = a$ and $r = 2a$

Let $x = r/a$. The polynomial is:

$$P(x) = -3x^3 + 7x^2 - 4$$

Using the provided hint, factor $P(x)$:

$$P(x) = -(x-1)(x-2)(3x+2)$$

The roots are:

$$x_1 = 1 \implies r_1 = a$$

$$x_2 = 2 \implies r_2 = 2a$$

$$x_3 = -2/3 \implies \text{unphysical } (r > 0)$$

In the physical domain $r \in [a, 2a]$, $(x-1) \geq 0$, $(2-x) \geq 0$, and $(3x+2) > 0$, so $\dot{r}^2 \geq 0$.

The particle comes to rest at $r = a$ and $r = 2a$, oscillating periodically between them. ■

Part (e): Cord Tension $T(r)$

Applying Newton's second law to P_2 with the vertical axis pointing up: the depth below the hole is $h = 2a - r$, so $z_2 = -h = r - 2a$ and $\ddot{z}_2 = \ddot{r}$; the tension acts upward and the weight downward:

$$m\ddot{z}_2 = T - mg \implies m\ddot{r} = T - mg \implies T = m(g + \ddot{r})$$

To find $\ddot{r}$, differentiate the energy equation $\dot{r}^2 = \frac{7}{3}ga - gr - \frac{4ga^3}{3r^2}$:

$$2\dot{r}\ddot{r} = -g\dot{r} + \frac{8ga^3}{3r^3}\dot{r} \implies \ddot{r} = -\frac{1}{2}g + \frac{4ga^3}{3r^3}$$

Substitute into T:

$$T = m\left[g + \left(-\frac{1}{2}g + \frac{4ga^3}{3r^3}\right)\right] = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$$

This is the formula requested in the problem statement. ■

Part (f): Equations with Friction Coefficient μ

On the rough horizontal plane, normal force is $N_1 = mg$.

Friction force: $\mathbf{F}_f = -\mu mg\frac{\mathbf{v}_1}{|\mathbf{v}_1|} = -\mu mg\frac{\dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta}{\sqrt{\dot{r}^2 + r^2\dot{\theta}^2}}$.

New coupled equations of motion:

$$2m\ddot{r} - mr\dot{\theta}^2 + mg = -\mu mg\frac{\dot{r}}{\sqrt{\dot{r}^2 + r^2\dot{\theta}^2}}$$

$$m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = -\mu mg\frac{r\dot{\theta}}{\sqrt{\dot{r}^2 + r^2\dot{\theta}^2}}$$

PHASE 4

Physical Interpretation & Dimensional Verification

- When $r = a$: $T = mg(1/2 + 4/3) = \frac{11}{6}mg > mg$, pulling P_2 upwards.
- When $r = 2a$: $T = mg(1/2 + 4/24) = mg(1/2 + 1/6) = \frac{2}{3}mg < mg$, letting P_2 accelerate downwards.
- The tension oscillates smoothly between $\frac{2}{3}mg$ and $\frac{11}{6}mg$, maintaining the periodic radial cycle.
- **Dimensional check:** $[\dot{r}^2] = \mathrm{m}^2/\mathrm{s}^2$ and $[g\,a^3/r^2] = [g\,r] = \mathrm{m}^2/\mathrm{s}^2$; $[T] = [m\,g] = \mathrm{N}$; the tension formula $T = mg\left(\frac{1}{2} + \frac{4a^3}{3r^3}\right)$ is homogeneous because a^3/r^3 is dimensionless.
- **Numerical check (RK4, step 10^{-4}, $g = a = 1$, $v_0^2 = 8/3$):** integrating $\ddot{r} = -\frac{1}{2} + \frac{4}{3r^3}$ from $r = 1, \dot{r} = 0$ gives $r_{min} = 1.0000000$ and $r_{max} = 2.0000000$ (the oscillation of part (d)), and the tension $T = mg(\frac{1}{2} + \frac{4}{3r^3})$ evaluated along the orbit ranges between

$0.666667\,mg$ at $r = 2a$ and $1.833333\,mg$ at $r = a$, equal to $\frac{2}{3}mg$ and $\frac{11}{6}mg$. The radial period is $6.3424\sqrt{a/g}$ (RK4, step 10^{-5}).

- **Limit (general launch speed):** differentiating the energy equation for an arbitrary v_0 gives $\ddot{r} = -\frac{1}{2}g + \dfrac{a^2 v_0^2}{2r^3}$ (for $v_0^2 = 8ga/3$ this is the $\ddot{r}$ of part (e)). For $v_0^2 = ga$ it vanishes at $r = a$, recovering the circular orbit of part (a) with $T = m(g + \ddot{r}) = mg$.

Source: Problems.pdf (Page 30)